

Welcome to the dbt MCP Hands on Lab!

While we're waiting for folks to join, feel free to get started on setup:

1. **Sign up for a workshop sandbox**
 - a. Important: Log out of dbt if you are in an existing account AND/OR open the workshop sandbox in an incognito window.
 - b. Sign up for the workshop (don't forget to save the username/password!):
 - i. URL: <http://workshops.us1.dbt.com/workshop>
 - ii. WORKSHOP: [Creating Context with the dbt MCP Server](#)
 - iii. PASSCODE: [Summit2026!](#)
 - c. Once you sign in, a [{user-name}@dbtlabsworkshop.com](#) will be created for you
 - i. Password: [1DataControlPlane!](#)
 - d. **Save both this username and password!**
2. **Run the initial job in the production environment: [dbt build](#)**

NOTE: If you don't have an AI client or don't have access to configuring MCPs, you can use a free Claude Desktop Account to follow the lab.

Creating Context with the dbt MCP Server

How to connect to and use the dbt MCP server in daily dbt work



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Agenda

- What is an MCP and why does it matter?
- Configure and verify your MCP connection
- Hands on scenario
 - Get your bearings
 - Verify the business question
 - Trace the blast radius
 - Write the incident summary
- Troubleshooting common errors
- Where to go from here



Section One

What is an MCP and why does it matter?

The Problem

What gaps exist in AI out of the box?

AI cannot automatically see:

- Your models, sources, and schemas
- Column names
- Lineage and downstream dependencies
- Metric definitions from the Semantic Layer
- Run history, job status, and freshness

The result? It sounds plausible but is likely wrong.

Plausible SQL against non-existent tables, answers that cannot be trusted, and no ability to take action.

What MCP is:

An open protocol that lets AI clients connect to tools and resources through one standard interface

01

One protocol, many tools

"USB-C for AI": build the integration once and any compliant client can use it

02

Open standard

Not tied to a single vendor or model, so both servers and clients are interchangeable

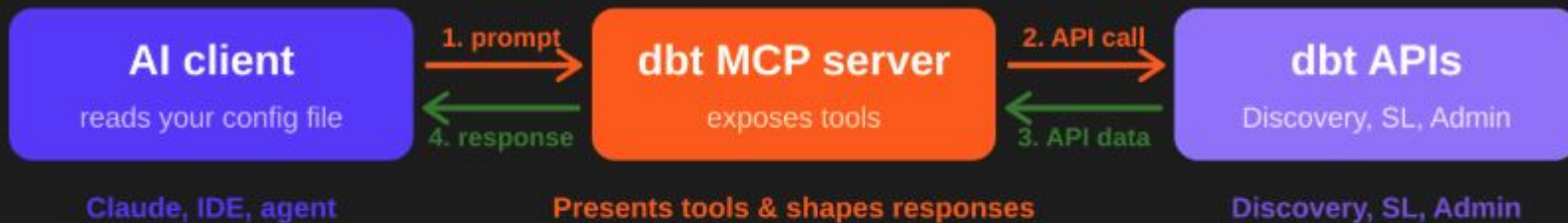
03

Reason + act

Tools let the client actually perform actions in and get context from external systems

How MCP Works

Request / response loop



Request

AI client calls a tool, and the MCP server makes the request to dbt platform APIs.

Response

APIs return data, the MCP server shapes it, the client processes it — loop until done.

Local vs. Remote

Similar tools, different deployment

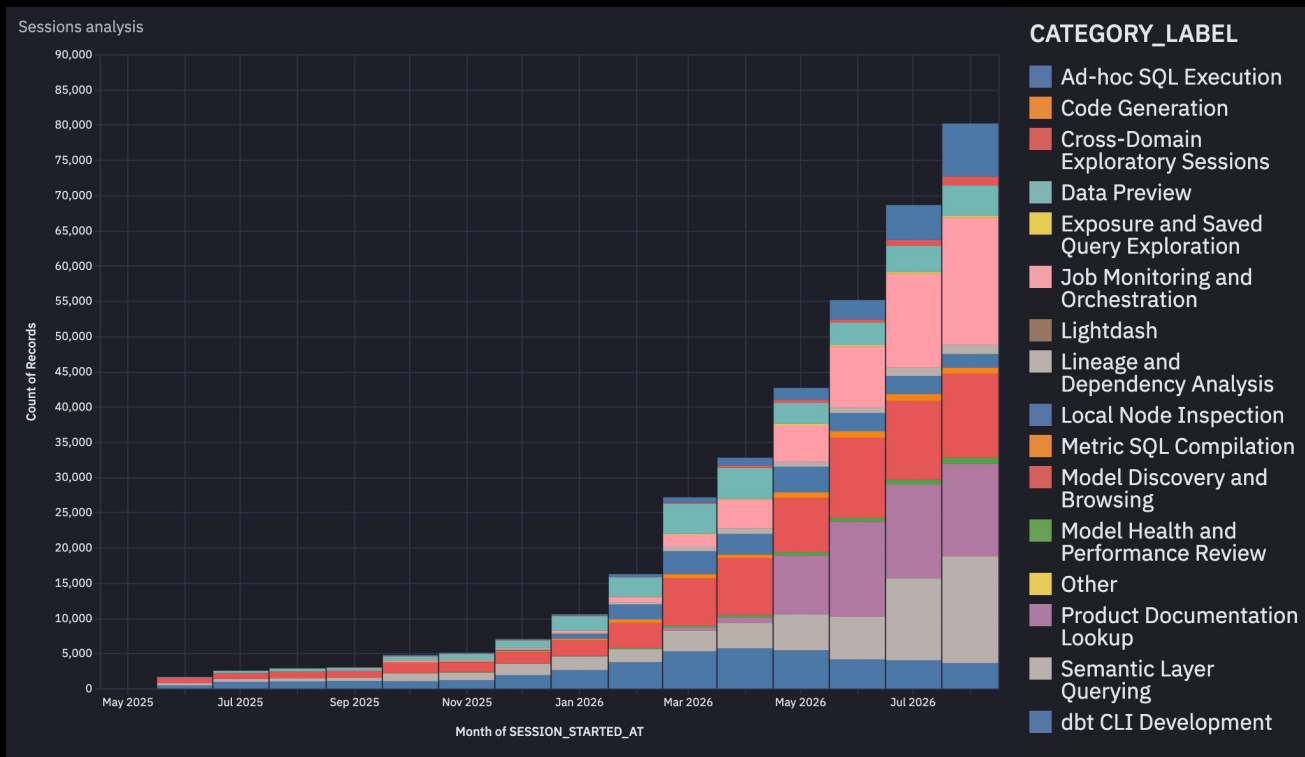
Local: *Generally* for when you're building and iterating (local dev)

- Server runs on your machine
- Runs beside your dbt project on your machine
- Fast loop for development, testing, and debugging
- Full local project context and CLI command tools
- Great for a single practitioner

Remote: More for when a team or agent wants to consume information

- Hosted with no local setup per user
- Centralized auth and governance
- Consistent access across the org
- Great for consumption of project context, dbt Semantic Layer, or interaction with the Discovery/Admin API

MCP for Context Consumption



Tool Categories

What the dbt MCP server exposes

NOTE: These are four consumption-specific tool sets but additional tool sets exist for SQL execution, dbt CLI, the LSP for Fusion, dbt codegen, and MCP metadata.

Semantic layer:

Query governed metrics by their real definitions, not guessed SQL

Discovery

Find models, sources, macros, and metadata, so the model knows what actually exists

Product Docs

Allowing your client to access, query, and read docs.getdbt.com natively

Admin API

Check run status, trigger jobs, and inspect job failures

Section 1 Recap

dbt MCP helps to turn “sounds right” into grounded, trustable results

- MCP connects AI to your tools and data through an open standard
- The dbt MCP has local and remote variants
 - Growing remote, consumption-based usage
- Has Discovery, Semantic Layer, Admin, Product Docs tool sets and more
- The payoff: grounded, actionable answers instead of confident guesses



Section Two

Configure and verify your MCP connection

Getting Started

1. Sign up for a workshop sandbox

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2. Run the initial job in the production environment: **dbt build**

Types of Authentication

Auth methods currently supported by dbt's remote MCP server

Personal Access Token

- Great for individual usage
- Fallback for clients that do not support OAuth

Service Token

- Great for usage in a deployed agent or BI tool
- Does not allow use of the `execute_sql` tool

OAuth

- Lowest-friction setup
- MCP client must support OAuth for HTTP servers

Configuration

Where does configuration live?

- Depends on your client:
 - .mcp.json
 - .vscode/mcp.json
 - .codex/config.toml
 - /Users/{your-user}/Library/Application Support/Claude/claude_desktop_config.json
 - Client Application UI
 - Claude Desktop
 - ChatGPT

Configuration - Files

```
{
  "mcpServers": {
    "dbt": {
      "command": "uvx",
      "args": ["dbt-mcp"],
      "env": {
        "DBT_HOST": "cloud.getdbt.com",
        "DBT_TOKEN": "your-token-here",
        "DBT_PROD_ENV_ID": "12345",
        "DBT_PROJECT_DIR": "/path/to/project",
        "DBT_PATH": "/path/to/dbt"
      }
    }
  }
}
```

```
{
  "mcpServers": {
    "dbt": {
      "type": "http",
      "url": "https://YOUR_DBT_HOST_URL/api/ai/v1/mcp/"
    }
  }
}
```

Configuration - Client App UI

Connect to a custom MCP

[Docs](#) [🔗](#)

Name

MCP server name

Type STDIO Streamable HTTP

Command to launch

openai-dev-mcp serve-sqlite

Arguments

+ Add argument

Environment variables

Key Value

+ Add environment variable

Environment variable passthrough

+ Add variable

Working directory

~/code

Save

Customize

[Skills](#) **Connectors** [Plugins](#) [Yours](#) [Discover](#)

← Connectors

< Back

Connect dbt

Enter the details below to connect Claude to dbt.

Server URL

https://

Find yours at [the dbt docs](#).

- ChatGPT MCP UI (left)
- Claude Desktop - dbt Connector (top)

URL: Finding your MCP URL

- Navigate to Account Settings
- Click on Account and scroll down to Access URLs
- Copy the MCP Endpoint URL found at the bottom of the list

Access URL ⓘ

<https://tk626.us1.dbt.com/> Copy

The base URL where you can access the UI and Admin API for this account.

Semantic Layer JDBC API URL

<jdbc:arrow-flight-sql://tk626.semantic-layer.us1.dbt.com:443> Copy

The base URL for requests to the Semantic Layer JDBC API for this account.

Semantic Layer GraphQL API URL

<https://tk626.semantic-layer.us1.dbt.com/apl/graphql> Copy

The base URL for requests to the Semantic Layer GraphQL API for this account.

Discovery API URL

<https://tk626.metadata.us1.dbt.com> Copy

The base URL for requests to the Discovery API for this account.

MCP Endpoint URL

<https://tk626.us1.dbt.com/apl/ai/v1/mcp> Copy

The MCP server endpoint for connecting external AI tools to this account.

Configuring Remote dbt MCP with Claude Desktop + ChatGPT

1. Claude Desktop: Set up with remote dbt MCP Server
 - a. Uses the Official dbt Connector in the Anthropic Marketplace
 - b. Customize → Connectors → Search "dbt" → Connect to Claude → Connect dbt → Enter Server URL → OAuth
2. ChatGPT Desktop (Custom MCP App)
 - a. Plugins → Gear Icon → MCPs → + Add Server → Enter Server Name/URL → Streamable HTTP → Save → Authenticate

*No cloning of GitHub repo needed for either of these options

Configuring Remote dbt MCP with Coding Agents and IDEs

- <https://github.com/dbt-labs/dbt-summit-26-mcp-server>
- If you are using Claude Code, Codex, Cursor, VSCode
 - Clone the repo
 - Find your MCP client configuration in [mcp-config/remote-oauth/](#)
 - Follow the instructions in [mcp-config/README.md](#)



Section Three

Our first day at Merlin & Co. Apothecaries!

You've just joined the data team. At 9am, a courier owl from the Guild of Accountants, sends you an urgent message: *"One of the revenue dashboard is broken - finance needs your help before the quarterly review (which is in 30 minutes...)."*

Work through it using your MCP-connected AI.

Exercise 1 - Get your bearings

Explore the project, understand what fct_orders does, how the data is structured, and how revenue makes its way from raw source tables to a number you can trust.

Explore:

- How is revenue calculated?
- What operational systems feed into fct_orders?
- What is the grain of fct_orders and fct_order_items?
- Where is region sourced from?
- What metrics are available?
- What downstream exposures (dashboards) are dependent on this project?

Exercise 2 - Verify the business question

One of the revenue dashboards is down, but finance still needs a number. Use the Semantic Layer to query revenue by region in natural language while the fix is in progress.

Ask:

- *What is total revenue in gold crowns by shop region for the last 90 days, ordered from highest to lowest?*
- *Break that down further by sales channel: in_store, courier_owl, and marketplace.*

Think: *If reporting for regional revenue looks fine, what is actually broken? (hint: Look at downstream dependencies for revenue reporting ie `fct_orders+`)*

Exercise 3 - Trace the blast radius

Before you fix anything, understand what else breaks

Ask:

- *What models and reports are affected? If this model fails or returns bad region data, who and what is affected?*
- *How can we fix the issue?*

Goal: Produce a short list of affected reports and stakeholders so you can set expectations before the fix goes out.

Exercise 4 - Write the incident summary

Pull context and draft a message to finance

Your message should include:

- **What happened:** which model failed and why (e.g., a timestamp parsing error in ordered_at, or a region conformance gap between Grimoire CRM and Alembic Ops shop data)
- **What data is affected:** which regions and date range are impacted
- **Interim workaround:** the Semantic Layer query from Exercise 2 gives trustworthy numbers right now
- **ETA:** when the fix will be in and the dashboard restored

Bonus Round!

- Via the dbt MCP, ask the LLM to run a custom job via the Admin API:
 - `dbt build --vars '{DBT_BONUS_ROUND: true}'`
- Then, ask your agent about the break
- Questions:
 - What tool(s) did your agent call?
 - Why did this run fail?
 - What is the important takeaway of this bonus round?

Recap - what did we do?

With the dbt MCP exposing a governed SL + project context to your agent:

1. Explored the project, transformation context, available metrics, and end-to-end lineage
2. Used the SL to query revenue metrics and triage what actually broke and what is affected
3. Compiled an incident summary for the finance team with full context on cause, impact, and the fix
4. Leveraged dbt MCP to diagnose a production run failure and saw how data tests can prevent this bugs in the future



Section Four

Troubleshooting common failures

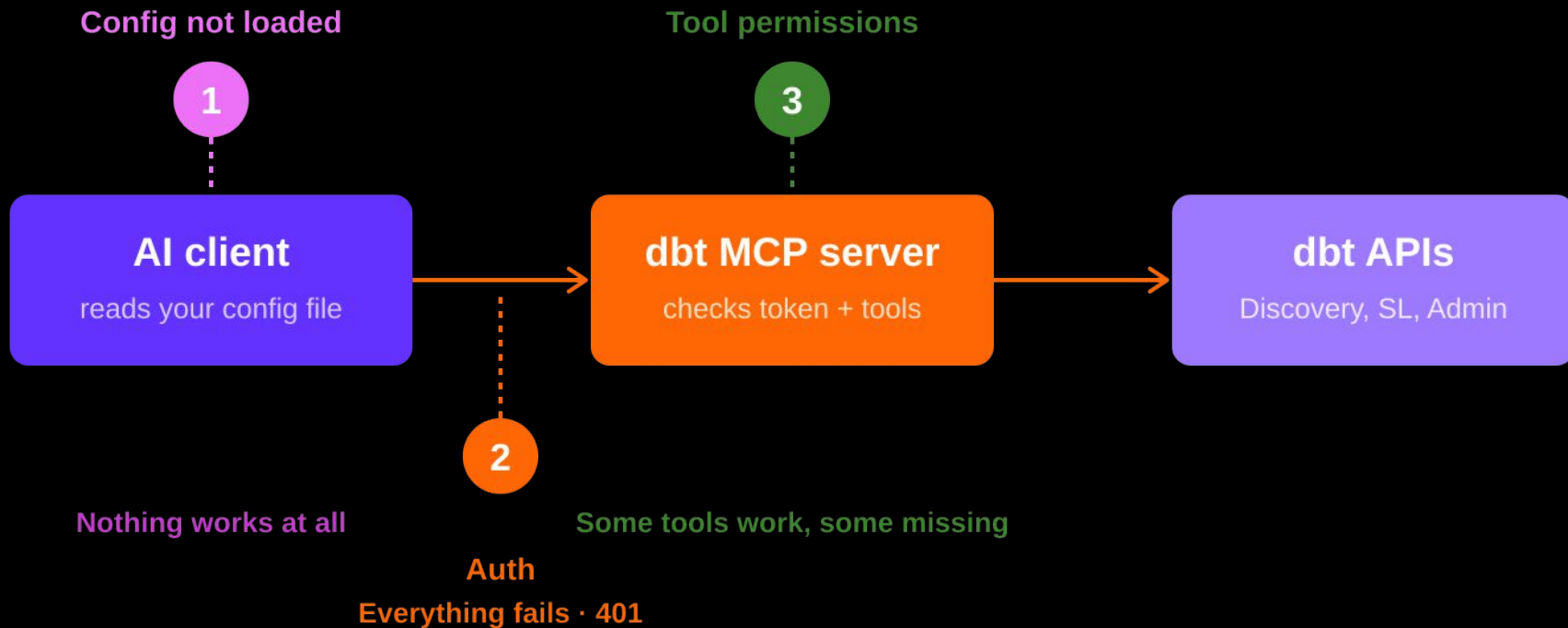
Troubleshooting common failures

Nearly every failure with the dbt MCP fits into one of three buckets

- Authorization: Token/Oauth issues when connecting to the MCP server
- Tool Permissions: The tools that can be used by your implementation of the MCP server
- Client Issues: Malformed JSON, unsupported protocols, etc.

Recognize the bucket, and you're already most of the way to a solution!

Where things break



Auth errors

Bad tokens, expired credentials, wrong headers, etc

Symptoms:

- 401/Unauthorized: Token is incorrect or missing the Token/Bearer prefix
- Worked, then stopped working: Expired or revoked access token
- "Forbidden" on some calls: Possible wrong header values

Fixes

- Re-copy the token fresh and paste
- Confirm the prefix (Token <token> or Bearer <token>)
- Check the header names being passed against dbt docs
- Confirm the IDs are just the numeric values
- If execute-sql doesn't work, make sure you are using a PAT or Oauth

Tool permissions

The server exposes several tool groups, not all can be used by the remote MCP

Symptoms:

- Agent says a given capability “isn’t available” despite a working connection
- Discovery works but execute-sql is absent
- Semantic layer queries fail while metadata works

Fixes

- Review the active tools available by the agent
 - /mcp in Codex and Claude Code
 - Tools & MCP panel in Cursor
 - MCP Indicator in Claude Desktop
- Review any missing tools and add
- If you are attempting to run dbt CLI commands, you will need local MCP

Client errors

I pasted the config and nothing happened!

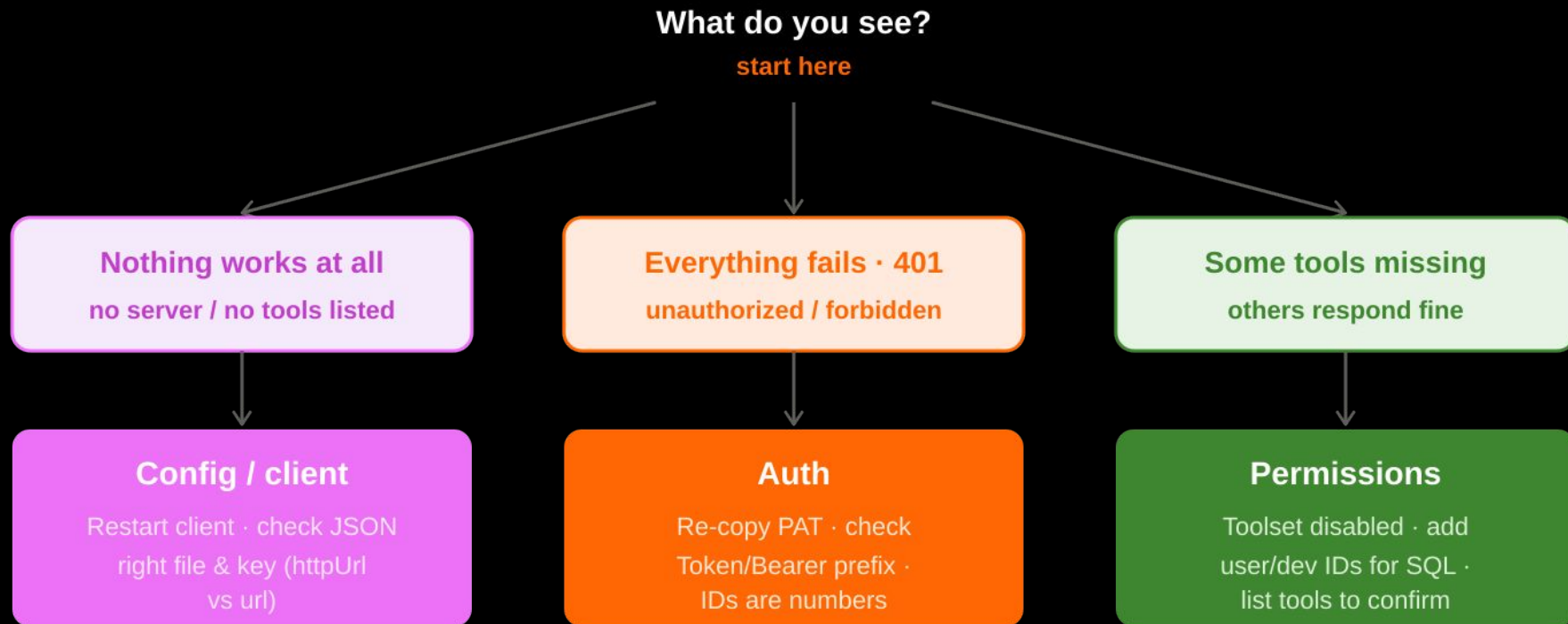
Symptoms:

- Error when restarting the client saying the MCP server could not be loaded
- Client showing not active MCP servers

Fixes

- Client restart (often required after a new MCP is added)
- Malformed JSON Check for trailing commas, missing braces.
- Check the transport key, some clients use httpurl instead of url

Decision Tree





Section Five

Where to go from here

Connect MCP to your own dbt Project!

- dbt Docs: [About dbt Model Context Protocol \(MCP\) Server](#)
 - Ask: What's your use case?
 - Choose whether to use local or remote
 - Configure dbt MCP w/ your dbt Platform project's values
- Start simple:
 - Ask about your DAG and project assets
 - Inspect production runs
 - Build + query your semantic layer
 - Verify code with product docs questions

Use cases for dbt MCP

- Conversational analytics on governed data
 - An agent that connects to the dbt MCP to answer questions about data assets, lineage, metrics, etc.
- A `/slash` command in your dbt project that calls tools to triage failures in CI or production
- Ad hoc debugging, exploring models/platform runs, or impact analysis via access to the Discovery API, Admin API, and Job Run Artifacts to aid with development
- Agent + human discovery of data assets/platform context
 - Faster onboarding and enrichment of critical business context
- Ground coding agents with product documentation
- Execute queries against your warehouse (*depending on data regulations)
- Operational agents
 - PR reviews, on-call, project health overviews, automatic ticket/issue filing w/ enriched context

What's shipped and where it's heading

Over the last 6 months:

- Token-efficiency improvements
 - Bounding tool call results and returning only what an agent needs
- Consolidating tools for a lighter context window footprint
- Calling and querying the dbt product docs directly
- Addition of `get_job_run_artifacts` to allow an LLM to parse, explore, and analyze raw dbt Artifacts produced by any step in a job run

What lies ahead:

- Search and context are expanding beyond structured metrics

Community resources

- Public Github: github.com/dbt-labs/dbt-mcp
 - CONTRIBUTOR guide
 - README setup, latest releases, and issues
- dbt Docs: <https://docs.getdbt.com>
 - Authoritative setup and usage instructions
- dbt Community Slack: #tools-dbt-mcp
 - Community support, maintainer discussion, and use case ideas
- Summit
 - Speak to fellow attendees and see how they are using the MCP server in their work!



Q&A

Ask the hosts any burning questions!

Thank you for joining!

We want to hear from you:

<https://dbtlearn.typeform.com/dbt-summit-labs>

